

Tech Note T26: Introduction to Many-Body Physics: Exploring Spin Models for Quantum Computing Scientists

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In this pedagogic note, we will introduce the fascinating realm of many-body physics to quantum information scientists through the exploration of spin models. Our focus will be on providing a beginner-friendly approach to understanding the concepts of Hamiltonians, energy levels, one-particle density matrices (OPDMs), and the effects of disorder. Think of this as a "Hello World" guide to gain an intuitive grasp of spin models, building upon the non-interacting Anderson model and primarily focusing on one-particle, one-dimensional problems. Whether you are a student or an enthusiast, we invite you to dive into the world of spins.

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I will assume you are familiar with qubits, spin, and some basic terms of many body systems. For background, see our lectures at the Qiskit Global Summer School, which are on the qiskit.org website and also on [youtube.com](https://www.youtube.com).

Spin models in many-body physics. Spin models play a crucial role in many-body physics. We use them to understand complex, collective behaviors of interacting particles, to devise simplified yet powerful frameworks to investigate fundamental physical principles, and to predict and design material properties. By helping us uncover insights and develop practical applications, spin models help drive scientific progress and guide technological advancements. So, let's get familiar with them.

26.1 Anderson model

In the Anderson model, particles (such as electrons) are assumed to hop between lattice sites in a regular lattice. For simplicity, let's work with a spin model. Imagine a one-dimensional lattice of 6 sites as depicted in Fig. 1. Each site has a spin that can have a bosonic spin excitation or no excitation on it. Since we are working with spins at each site, we can have either zero or one excitations per site. The lattice can either have open or closed (i.e. periodic) boundary conditions.

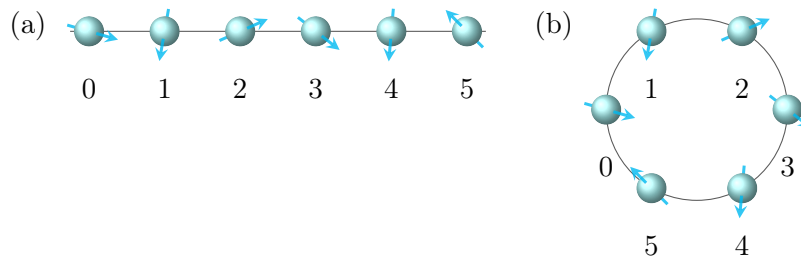
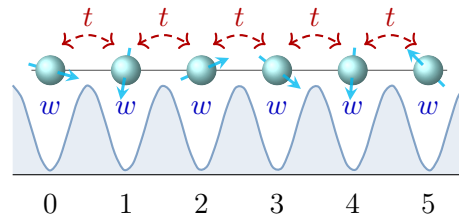


Figure 1: **One-dimensional (1D) Anderson spin model.** The illustration portrays a one-dimensional spin chain with both open (top) and closed (bottom; meaning, periodic) boundary conditions for $L = 6$ spins. Each site is affected by a specific potential, defined by the on-site energy w_i at site i . In this model, the interaction coupling $t_{i,i+1}$ is presumed to involve only the nearest neighbors. We can at first assume translational invariance for all coefficients, $w_i = w$ and $t_{i,i+1} = t$.

The Hamiltonian of this one-particle hopping with on-site potential model is

$$H = -\frac{t}{2} \sum_{i=0}^{L-1} \left(c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i \right) + w \sum_{i=0}^{L-1} \left(c_i^\dagger c_i - \frac{1}{2} \hat{I} \right) \quad (1)$$

for L sites (spins). The index of the first sum runs to $L - 2$ for open boundary conditions and to $L - 1$ for closed, with the understanding that all indices are modulo L . The hopping or tunneling rate is t and w is the on-site energy. Notice that the problem is translationally invariant the the periodic boundary conditions. Otherwise, it is almost translationally invariant except for the edges. How can we think of the origin of this Hamiltonian? How can we depict this system visually? Here is one way to think about it visually:



We have illustrated a spin chain system with open boundary conditions here. The red arrows indicate the nearest neighbor hopping, controlled by the parameter t . The on-site energies, represented by blue marks below, are uniform and dictated by the parameter w . A useful perspective for understanding both the hopping term and the on-site energy is to envision particles moving within a specific potential energy landscape, for instance, the periodic sine wave depicted here. Disorder in the potential could introduce disorder in the potential energy at each site w_i (and perhaps the hopping terms as well).

How do we begin to understand the properties and dynamics of the system? For smaller system sizes, we can undertake a numerical diagonalization to examine the model. To simplify the process, let's begin with distinct hard-core bosons — essentially, just spins. In this approach, we avoid the complexities of symmetrizing particles for indistinguishable bosons or fermions, making the 'c' operators easier to handle directly, akin to how we interact with qubits.¹

26.2 Anderson model with spins (hard-core bosons)

Consider particle which are distinguishable spin-1/2 bosons. In terms of qubit Pauli operator on each site i , the single particle operators are

$$c_i = \frac{1}{2} (X_i + iY_i) = |0\rangle \langle 1|_i, \quad (2)$$

where the index i labels the spatial index of the spins. Keep in mind X_i is an operator in a 2^L dimensional Hilbert space. Otherwise, these are the standard qubit Pauli operators.²

The on-site energy term is a scaled Pauli Z operator,

$$c_i^\dagger c_i = \frac{1}{2} (I - Z) = |1\rangle \langle 1|_i = \begin{matrix} \langle 0| & \langle 1| \\ \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{matrix}, \text{ and}$$

$$c_i^\dagger c_i - \frac{1}{2}I = -\frac{1}{2}Z = \frac{1}{2} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

For a brief moment, let's call this last term $\hat{W}_i := w \left(c_i^\dagger c_i - \frac{1}{2}I \right)$. Thus, the zero state $|0\rangle_i$ has a lower energy expectation value with respect to this Hamiltonian term. In particular, $\hat{W}_i |0\rangle_i = -\frac{1}{2}w |0\rangle_i$ and $\hat{W}_i |1\rangle_i = +\frac{1}{2}w |1\rangle_i$. These are eigenvectors and eigenvalues. If we assume, $w > 0$ and $t = 0$, then the ground state is $|0\rangle$.³

¹The one particle (defined in the spin spatial basis) occupation or density corresponds to the operator $n_i := c_i^\dagger c_i$.

²Notational convention: Sometimes, you will see $c_i = \frac{1}{2} (X_i - iY_i) = |1\rangle \langle 0|_i$ in which case the ground state is thought of as the $|1\rangle$ state. Then, $c_i^\dagger c_i = \frac{1}{2} (I - Z) = |0\rangle \langle 0|_i$ and $c_i^\dagger c_i - \frac{1}{2}I = \frac{1}{2}Z$.

³By the way, here we have been loose in our discussion, since we only looked at the single particle (trace out the rest of the system if you want or make $L = 1$). We will remedy soon.

The interaction term is proportional to

$$c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i = X_i X_{i+1} + Y_i Y_{i+1} = \begin{matrix} & \langle 00| & \langle 01| & \langle 10| & \langle 11| \\ \begin{matrix} |00\rangle \\ |01\rangle \\ |10\rangle \\ |11\rangle \end{matrix} & \begin{pmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & 0 \end{pmatrix} \end{matrix},$$

which can be seen to be single particle number conserving since it does not change the parity of the two-qubit reduced blocks. The number of up and down spins in the blocks stay fixed.

26.2.1 Simple regime 1: Hopping-free, translationally-invariant, periodic-BC regime

26.2.1.1 Hamiltonian, eigenvalues (spectrum), and eigenvectors

Let's first get comfortable with the pictures we will use to analyze the more complex system by starting with the simplest one first that we can analyze analytically (since it's nearly trivial) and we can easily plot numerically for small system sizes. Hence, consider the Hamiltonian of Eq. (1) with $t = 0$,

$$H = w \sum_{i=0}^{L-1} \left(c_i^\dagger c_i - \frac{1}{2} \hat{I} \right) = -w \sum_{i=0}^{L-1} Z_i .$$

It's easy to see what the eigenvectors, eigenvalues, and conserved quantities and quasi-particles are, since there are no interactions in the system. In other words, it's not really a many body system as much as a collection of a bunch of single systems, a perfect place to start. Let's check the basis state $|b\rangle = |b_{L-1} \cdots b_1 b_0\rangle$ with the L -bit binary string b ; i.e., $b \in \mathbb{Z}_2^L$, with $b_i \in \{0, 1\}$ for $i = 0, \dots, L-1$. For the qubit ordering, let's assume $|b_{L-1} \cdots b_1 b_0\rangle = |b_{L-1}\rangle_{Q(L-1)} \otimes \cdots \otimes |b_0\rangle_{Q1} \otimes |b_0\rangle_{Q0}$, where qubit 0 is given by the right-most bit-string.⁴

$$\begin{aligned} H |b\rangle &= -w \sum_{i=0}^{L-1} Z_i |b_{L-1} \cdots b_1 b_0\rangle \\ &= E(b) |b\rangle , \\ E(b) &= -w \sum_{i=0}^{L-1} (-1)^{b_i} . \end{aligned}$$

In other words, the binary-string ket $|b\rangle$ is an eigenvector, and it's energy $E(b)$ is just a measure of the Hamming weight of the bit-string — in other words, the energy just depends on how many ones and zeros there are. The more ones, the higher the energy. We could interpret this as how many particles are in the system. Because of the translational invariance of H , it doesn't matter spin is up (or which site the single particle is on). This is seen reflected in the Hamming block plots in Fig. 2.⁵

Note that the term $c_i^\dagger c_{i+1} + c_{i+1}^\dagger c_i$ won't change the Hamming weight of the eigenvector, hence the eigenvectors will only mix inside a block, as can be seen in Fig. (7)(a).

⁴With this convention, X on qubit 0 is $X_0 = I_{[L-1]} \otimes I_{[L-2]} \otimes \cdots \otimes X_{[0]}$, and the Hilbert space structure is $\mathcal{H} = \mathcal{H}_{Q_{L-1}} \otimes \cdots \otimes \mathcal{H}_{Q_1} \otimes \mathcal{H}_{Q_0}$.

⁵In numerical practice for creating the plot, I played a trick to sort the eigenvalues in a nice way. I added small linear gradient to make the $i = 0$ qubit the lowest energy one, then the $i = 1$ qubit the second lowest energy particle, and so on.

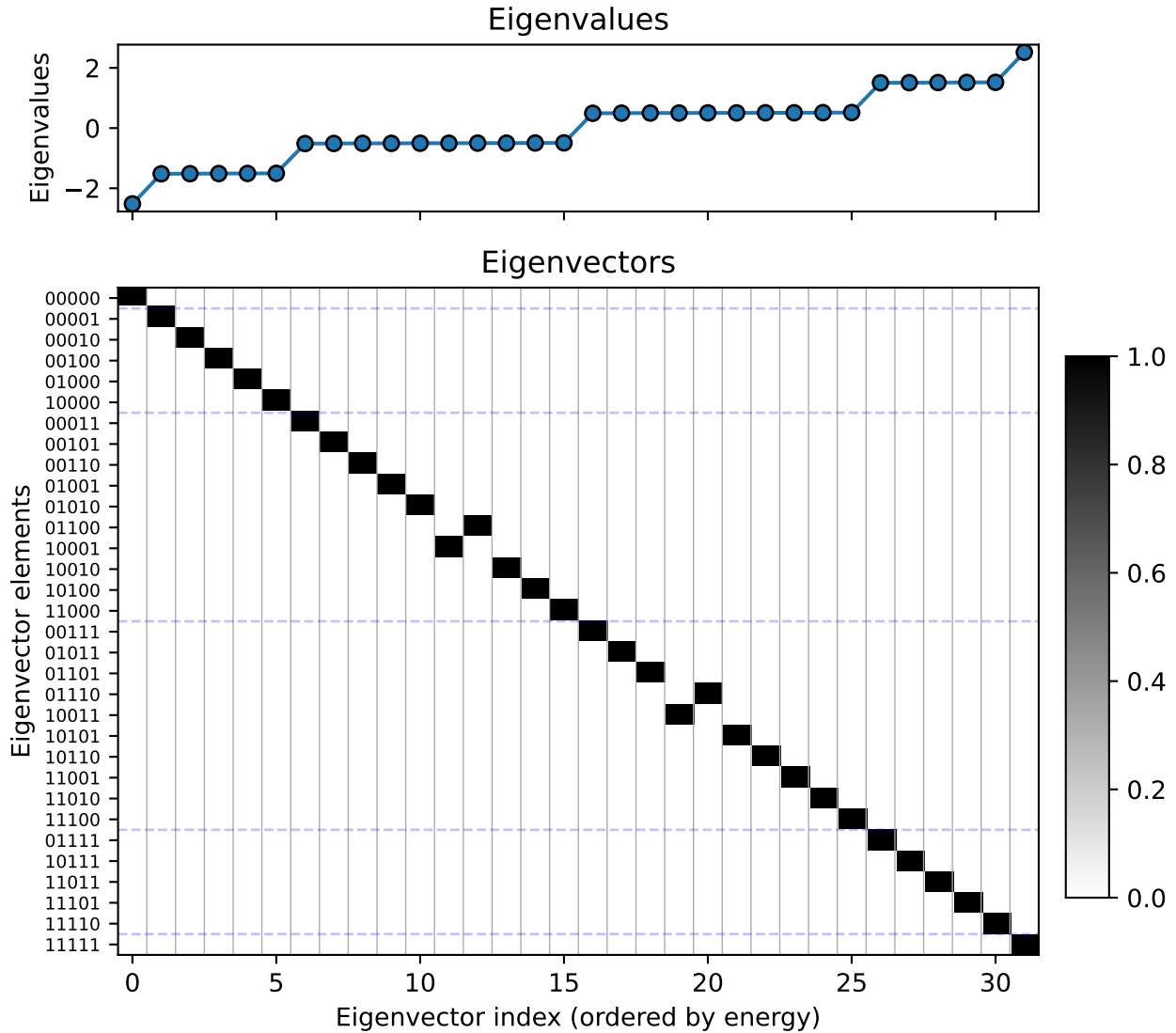


Figure 2: Numerical simulation for the hopping-free model of Sec. 26.2.1 for $L = 5$ spins, with periodic boundary conditions, and with $w = 1$ and $t = 0$. **(Top panel)** **(Lower panel)** The many-body wavefunctions of the system (eigenvectors). Plotted is the The bit-strings of the bare basis are sorted according to Hamming weight blocks first (indicted by the horizontal lines) and then lexicographically. There are $L + 1 = 6$ Hamming weight blocks. Note that $|000001\rangle$ is a state with one excitation on the zeroth qubit Q_0 .

26.2.1.2 Visualizing the eigenfunctions: particle density

Density visualization of the system. The particle site density $\langle n(i) \rangle := \langle c_i^\dagger c_i \rangle = |1\rangle \langle 1|_i$ for the one-particle occupied eigenstates for the $L = 5$ chain (i.e., eigenvectors 1 – 5) are as expected, $\langle n(i) \rangle = \delta_{i(k-1)}$ for the k -th eigenvector as shown in Fig. 3. Here, this is an excellent representation because the eigenvectors are eigenfunctions of this operator, and the Hamiltonian commutes with it.

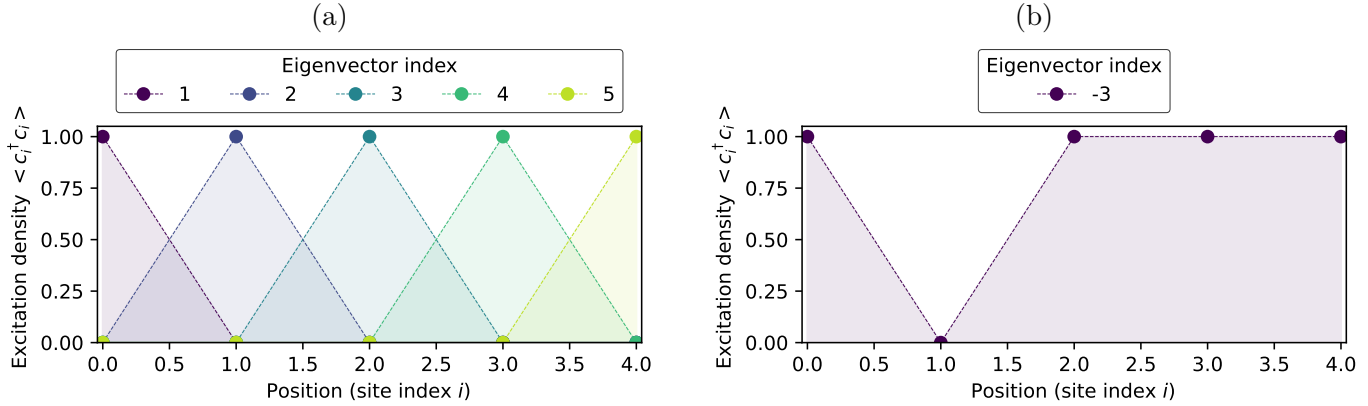


Figure 3: Density visualization of the for the density $\langle n(i) \rangle := \langle c_i^\dagger c_i \rangle$ (a) Hamming weight 1 modes, which look like a single particle localized at a site i and (b) for one of the highest eigenvectors (-3), which looks like a hole excitation from the all-filled state. This is related to the Jordan-Wigner mapping.

Eigenvectors above half-energy (i.e., above zero energy) look like they have holes instead of particles, see panel (b).

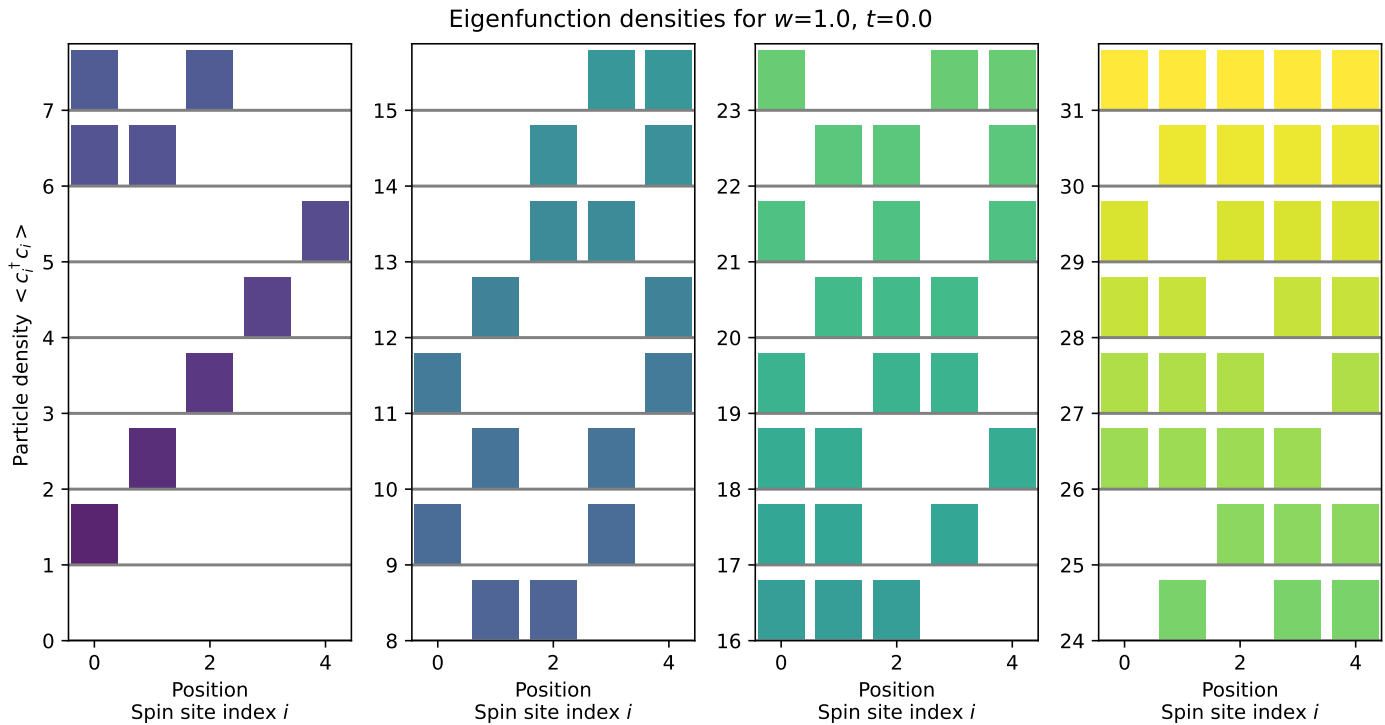
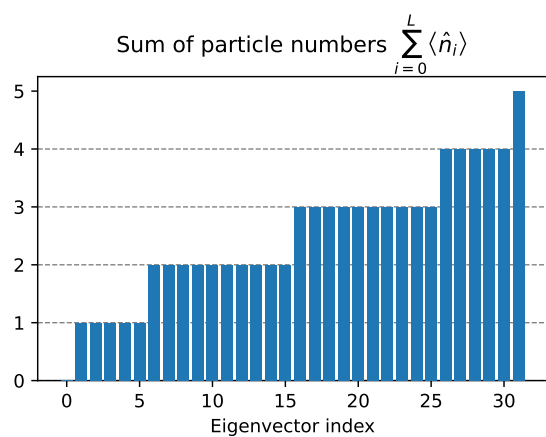


Figure 4: **Density visualization** of the for the density $\langle n(i) \rangle := \langle c_i^\dagger c_i \rangle$ for all eigenvectors. Note, rescaled from 1 maximum to 0.8 for visual clarity.

The number of simple particles:



26.2.1.3 Visualizing the eigenfunctions: spins

Spin visualization of the system. In Fig. 5, we plot the one-particle spin vectors $\langle \vec{S}_i \rangle := (\langle \hat{X}_i \rangle, \langle \hat{Y}_i \rangle, \langle \hat{Z}_i \rangle)$ for an example eigenstate. The global magnetization is defined as the average over the single magnetizations

$$\langle \vec{S} \rangle := \frac{1}{L} \sum_{i=0}^{L-1} \langle \vec{S}_i \rangle = \frac{1}{L} \sum_{i=0}^{L-1} (\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle) .$$

This representation will work well when there isn't much entanglement and correlations. When the spins are entangled, we will see that the individual spin vectors will tend to decay away, as how for a Bell pair, the one particle observables are all zero, but the correlators (higher weight Paulis) have maximum definite values. It is not easy to visualize these higher weight Paulis easily, since there are exponentially many. Later, we will look at the LIOMs and Pauli densities as one way to map them out.

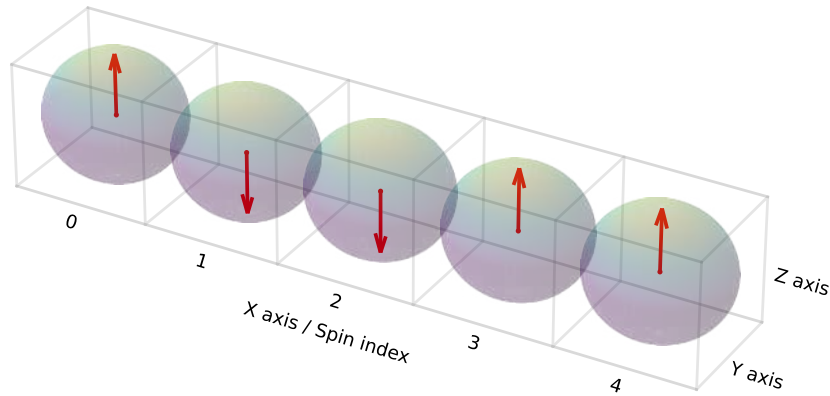


Figure 5: Illustration of the one-particle spin vectors $\langle \vec{S}_i \rangle = (\langle \hat{X}_i \rangle, \langle \hat{Y}_i \rangle, \langle \hat{Z}_i \rangle)$. Here, the 9th eigenvector (index 8) is plotted for $w = 1$ and $t = 0$, corresponding to $|00110\rangle \equiv |8\rangle$. Recall the left-most spin corresponds to the right-most bit in the binary encoding. Keep in mind that the spins pointing up mean $\langle Z \rangle = -1$. In other words, $|0\rangle \equiv |\uparrow\rangle$ and $|1\rangle \equiv |\downarrow\rangle$; hence, $|00110\rangle = |\uparrow\uparrow\downarrow\downarrow\uparrow\rangle$, which when written out as a spin configuration over the spin chain with Q_0 on the left (as in the picture above) is the reverse of $\uparrow\uparrow\downarrow\downarrow\uparrow$. The global magnetization is $\langle \vec{S} \rangle = \frac{1}{L} \sum_{i=0}^{L-1} (\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle) = (0, 0, 1/5)$.

26.2.2 One-particle density matrix (OPDM)

The SPDM,

$$\rho_{ij} := \langle c_i^\dagger c_j \rangle - \langle c_i^\dagger \rangle \langle c_j \rangle ,$$

for each of the eigenstates is trivial for $t = 0$; it's just a diagonal matrix with the binary string label of the eigenvector on the diagonal, i.e.,

$$\rho_{ij}(|b\rangle) = \text{Diag}(b) .$$

And its eigenvectors are those of the Hamiltonian, with proportional energies. Note that the SPDM ρ is a $L \times L$ matrix, whereas the Hamiltonian H is $2^L \times 2^L$ matrix. However, for this problem, the SPDM tells us all there is to know about H , since it's so sparse and a single particle problem altogether.

The sum of the SPDM eigenvalues tells us how many single particles are in the system.

26.2.3 Simple regime 2: Hopping only

TODO:

Occupation spectrum (OPDM eigenvalues) vs. eigenvector

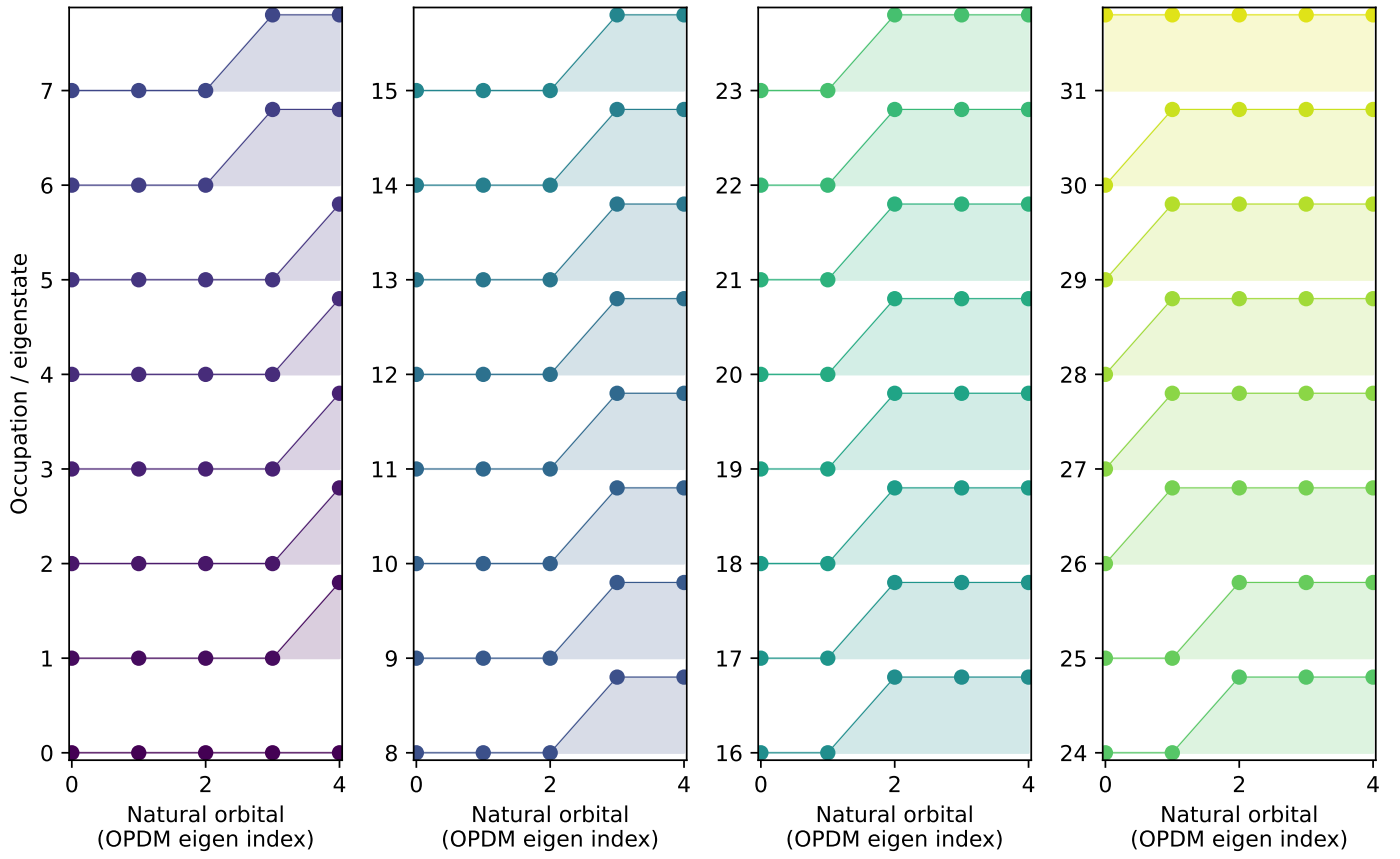


Figure 6: OPDM spectrum (eigspectrum) for $w = 1$ and $t = 0$. Note the offset for each eigenvector by its integer index, and the scaling of each curve by 0.8 for visual clarity.

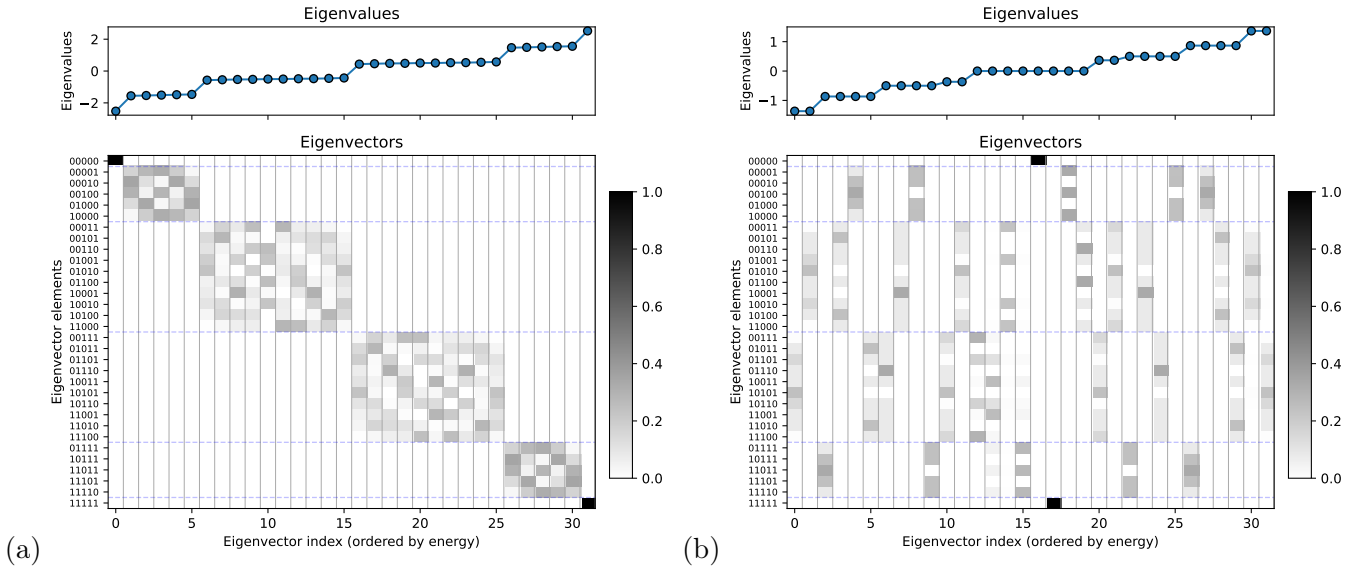


Figure 7: (a) $t = 0.05, w = 1$. Note the Hamming-block split and only mixing within a single Hamming block introduced by the hopping terms. (b) $t = 1, w = 0$ Note the symmetric split of the spectrum.

Occupation spectrum (OPDM eigenvalues) vs. eigenvector

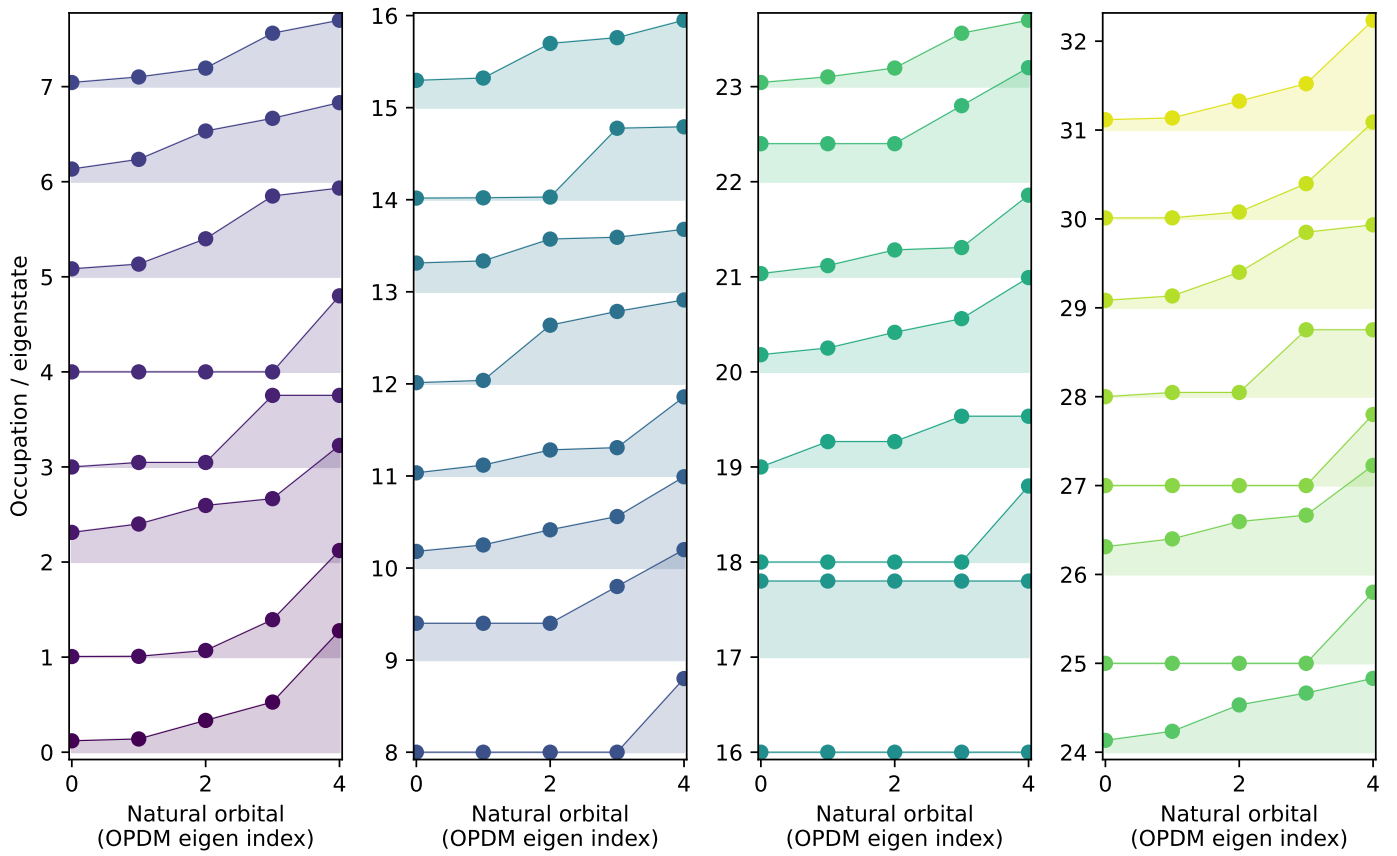


Figure 8: OPDM spectrum (eigspectrum) for $w = 0$ and $t = 1$. Note the offset for each eigenvector by its integer index, and the scaling of each curve by 0.8 for visual clarity.

- disorder
 - ergodic vs mbl, area / volume law?

26.3 Example integrals of motion in a toy model

Number of qubits n , since I use L here for the LIOM.

$$H = \sum_{i=0}^{n-1} c_i Z_i + \sum_{i \neq j} c_{ij} Z_i Z_j$$

$$[U, L] = 0$$

$$[U, L] = 0$$

$$L = \sum_{\mu=1}^{4^n-1} a_{\mu} P_{\mu}$$

$$\langle L \rangle = \text{const}$$

$$L_0 = Z_0$$

$$[H, Z_0] = 0 \quad [H, Z_1] = 0 \quad \dots$$

$$L_0 = Z_0, L_1 = Z_1, \dots$$

$$|l_0 l_1 \dots l_{n-1}\rangle = |l_0\rangle_{L_0} \otimes |l_1\rangle_{L_1} \otimes \dots \otimes |l_{n-1}\rangle_{L_{n-1}}$$



$$L_k = \sum_{\mu \in \mathcal{N}(k)} a_{\mu} P_{\mu}$$

References:

- “One-particle density matrix characterization of many-body localization”